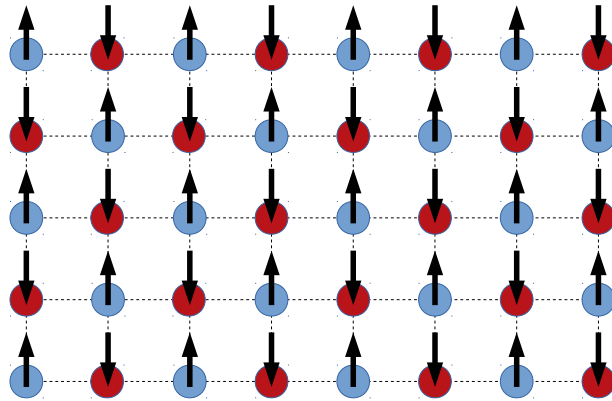


PHYS481: Advanced Magnetism

Lectures 8–10: Heisenberg Model



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Need for Quantum Theory to Describe Coupling Between Magnetic Moments

- Interaction energy of a pair of magnetic dipoles with magnetic moments $\mathbf{m}_1$ and $\mathbf{m}_2$, separated by the vector $\mathbf{R}$:

$$V_{\text{int}} = \frac{\mu_0}{4\pi} \left[\frac{\mathbf{m}_1 \cdot \mathbf{m}_2}{R^3} - \frac{3}{R^5} (\mathbf{m}_1 \cdot \mathbf{R})(\mathbf{m}_2 \cdot \mathbf{R}) \right].$$

- Using $|\mathbf{m}_1| = |\mathbf{m}_2| \sim \mu_B$ and $|\mathbf{R}| \sim 1 \text{ \AA}$ gives

$$V_{\text{int}} \sim \mu_0 |\mathbf{m}_1| |\mathbf{m}_2| / (4\pi |\mathbf{R}|^3) \sim 10^{-23} \text{ J},$$

which is equivalent to a temperature of $\sim 1 \text{ K}$.

- Magnetic interaction is not strong enough to explain observed magnetic behaviour.
 - Many materials order below much higher temperatures (e.g., the Curie temperature of iron is $\sim 1000 \text{ K}$).
 - **There must be some other (nonmagnetic!) mechanism by which magnetic moments interact.**

Two-Particle Wave Functions

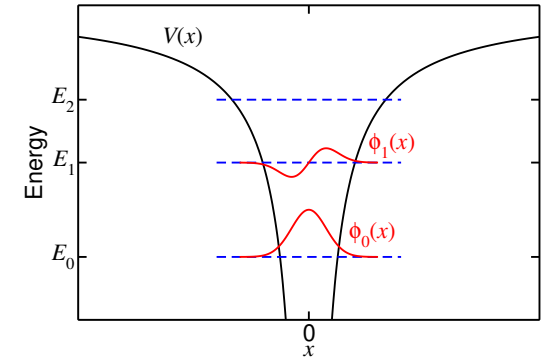
- Two-particle wave functions: $\Psi(\xi_1, \xi_2)$, where $\xi \equiv \{\mathbf{r}, \sigma\}$.
- Indistinguishability requires that $|\Psi(\xi_1, \xi_2)|^2 = |\Psi(\xi_2, \xi_1)|^2$.
 - Bosons: $\Psi(\xi_1, \xi_2) = \Psi(\xi_2, \xi_1)$.
 - Fermions: $\Psi(\xi_1, \xi_2) = -\Psi(\xi_2, \xi_1)$.
- Pauli repulsion: for two indistinguishable fermions

$$\Psi(\xi, \xi) = 0.$$

- Question: “How do we construct antisymmetric wave functions for the ground state of two noninteracting fermions?”

Singlet and Triplet Wave Functions (I)

- Suppose a particular spatial orbital is nondegenerate.



- Then there are two corresponding degenerate spin orbitals for a spin- $\frac{1}{2}$ particle: $\phi_0(\mathbf{r}) |\uparrow\rangle$ and $\phi_0(\mathbf{r}) |\downarrow\rangle$.
 - Spin-singlet wave function for two noninteracting particles

$$\Psi_S(\xi_1, \xi_2) = \phi_0(\mathbf{r}_1)\phi_0(\mathbf{r}_2) \frac{|\uparrow_1\rangle |\downarrow_2\rangle - |\downarrow_1\rangle |\uparrow_2\rangle}{\sqrt{2}}.$$

- Its energy is $E_S = 2\mathcal{E}_0$ and total spin quantum number is $S = 0$.
- More generally, for singlet wave functions:
 - spatial part is symmetric;
 - spin part is antisymmetric.

Singlet and Triplet Wave Functions (II)

- Now suppose the single-particle spatial orbital is doubly degenerate: $\phi_a(\mathbf{r})$ and $\phi_b(\mathbf{r})$ both have energy $\mathcal{E}_0$.
 - For two noninteracting fermions, there are four degenerate spinor orbitals to occupy:

$$\phi_a(\mathbf{r}) |\uparrow\rangle, \quad \phi_a(\mathbf{r}) |\downarrow\rangle, \quad \phi_b(\mathbf{r}) |\uparrow\rangle \quad \text{and} \quad \phi_b(\mathbf{r}) |\downarrow\rangle.$$

- There are $\binom{4}{2} = 6$ possibilities.
 - Again, we can form antisymmetric wave functions as products of space and spin parts.

Singlet and Triplet Wave Functions (III)

- Three linearly independent symmetric spatial wave functions:

$$\Phi_S(\mathbf{r}_1, \mathbf{r}_2) = \frac{\phi_a(\mathbf{r}_1)\phi_b(\mathbf{r}_2) + \phi_a(\mathbf{r}_2)\phi_b(\mathbf{r}_1)}{\sqrt{2}},$$

$$\Phi'_S(\mathbf{r}_1, \mathbf{r}_2) = \phi_a(\mathbf{r}_1)\phi_a(\mathbf{r}_2),$$

$$\Phi''_S(\mathbf{r}_1, \mathbf{r}_2) = \phi_b(\mathbf{r}_1)\phi_b(\mathbf{r}_2).$$

- Hence three singlet wave functions (energy = $2\mathcal{E}_0$, total spin $S = 0$):

$$\Psi_S(\xi_1, \xi_2) = \Phi_S(\mathbf{r}_1, \mathbf{r}_2) \frac{|\uparrow_1\rangle |\downarrow_2\rangle - |\downarrow_1\rangle |\uparrow_2\rangle}{\sqrt{2}},$$

$$\Psi'_S(\xi_1, \xi_2) = \Phi'_S(\mathbf{r}_1, \mathbf{r}_2) \frac{|\uparrow_1\rangle |\downarrow_2\rangle - |\downarrow_1\rangle |\uparrow_2\rangle}{\sqrt{2}},$$

$$\Psi''_S(\xi_1, \xi_2) = \Phi''_S(\mathbf{r}_1, \mathbf{r}_2) \frac{|\uparrow_1\rangle |\downarrow_2\rangle - |\downarrow_1\rangle |\uparrow_2\rangle}{\sqrt{2}}.$$

Singlet and Triplet Wave Functions (IV)

- A single antisymmetric spatial wave function:

$$\Phi_T(\mathbf{r}_1, \mathbf{r}_2) = \frac{\phi_a(\mathbf{r}_1)\phi_b(\mathbf{r}_2) - \phi_a(\mathbf{r}_2)\phi_b(\mathbf{r}_1)}{\sqrt{2}}$$

- Hence three triplet wave functions (energy = $2\mathcal{E}_0$, total spin $S = 1$):

$$\Psi_T^\uparrow(\xi_1, \xi_2) = \Phi_T(\mathbf{r}_1, \mathbf{r}_2) |\uparrow_1\rangle |\uparrow_2\rangle ,$$

$$\Psi_T^\downarrow(\xi_1, \xi_2) = \Phi_T(\mathbf{r}_1, \mathbf{r}_2) |\downarrow_1\rangle |\downarrow_2\rangle ,$$

$$\Psi_T^\otimes(\xi_1, \xi_2) = \Phi_T(\mathbf{r}_1, \mathbf{r}_2) \frac{|\uparrow_1\rangle |\downarrow_2\rangle + |\downarrow_1\rangle |\uparrow_2\rangle}{\sqrt{2}} .$$

Singlet and Triplet Wave Functions (IV)

Quantity	Singlet	Triplet
Total energy E for <i>nonint. particles</i>	$2\mathcal{E}_0$	$2\mathcal{E}_0$
Total spin S	0	1
Spatial part	Symmetric	Antisymmetric
Spin part	Antisymmetric	Symmetric
Contact pair density $ \Phi(\mathbf{r}, \mathbf{r}) ^2$	Nonzero	0

- For noninteracting particles, singlet and triplet states formed from degenerate spatial orbitals have the same energy.
- This changes once we take interactions into account:
 - In the triplet state, the contact pair density (i.e., probability density that two particles are in the same place) is zero.
 - This reduces the repulsive Coulomb interaction between particles.
 - Interactions favour triplet states.

Singlet and Triplet States for Electrons; Intra-atomic Exchange

- Exchange coupling constant (confusingly, symbol is $\mathcal{J}$, despite not being ang. mom.):

$$\begin{aligned}\mathcal{J} &= (E_S - E_T) / 2 \\ &= \frac{1}{2} \iint \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|} [|\Phi_S(\mathbf{r}_1, \mathbf{r}_2)|^2 - |\Phi_T(\mathbf{r}_1, \mathbf{r}_2)|^2] d^3\mathbf{r}_1 d^3\mathbf{r}_2 \\ &\approx \iint \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|} \phi_a(\mathbf{r}_1) \phi_b(\mathbf{r}_1) \phi_a(\mathbf{r}_2) \phi_b(\mathbf{r}_2) d^3\mathbf{r}_1 d^3\mathbf{r}_2,\end{aligned}$$

where the approximation is the neglect of electron correlation, i.e., the use of the wave-function form for noninteracting particles.

- Intra-atomic exchange: $\mathcal{J}^{\text{intra}} \sim +e^2/(4\pi\epsilon_0 a_B) \sim \text{eV}$.
- Hund's rule of maximum multiplicity: electrons in incomplete atomic subshells maximise total spin.
 - This is why d - and f -subshells of transition metals carry large spins and hence have large magnetic moments. N.b., unpaired electrons need not be involved in bonding in transition metals.

Aside: Hund's Rules for Angular Momenta

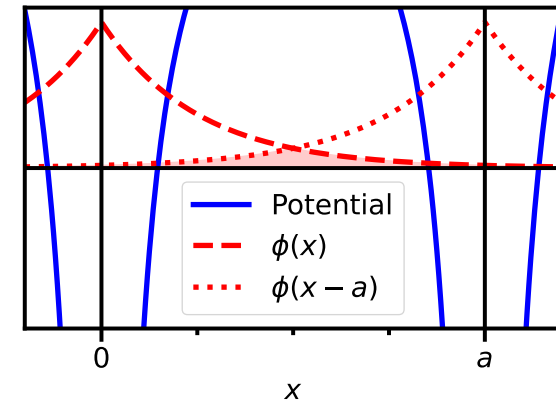
- **Closed shells:** totally antisymmetric combination of occupied states in a closed shell has $S = 0$ and $L = 0$. **Magnetic behaviour due to partially filled shells.**
- **Hund's rules** provide a qualitative summary of the nontrivial energetic orderings predicted by electronic-structure calculations and observed experimentally.

Consider an isolated atom with a given electronic configuration [e.g., $1s^2 2s^2 2p^2$]:

1. *Hund's Rule of Maximum Multiplicity.* The lowest-energy term $^{2S+1}L_J$ is such that the total spin quantum number S is maximised (equivalently, the spin multiplicity $2S + 1$ is maximised). Instance of general principle that Pauli exclusion keeps same-spin electrons spatially separated, reducing their Coulomb repulsion energy.
2. If there are still competing terms after application of Hund's first rule, the term with the largest total orbital angular momentum L is favoured energetically. Electrons orbiting in the same direction encounter each other less frequently.
3. The value of the total angular momentum quantum number is $J = |L - S|$ when the subshell is less than half full, and it is $J = L + S$ when the subshell is more than half full. This maximises the lowering of the energy due to spin-orbit coupling. If the subshell is exactly half full then Hund's first rule implies $L = 0$, so that $J = S$.

Interatomic Exchange in Solids (I)

- Assume neighbouring lattice sites (separation a), equal energy levels, and s -shell electrons.
- In the rest of these notes, “ $\mathcal{J}$ ” denotes exchange coupling constant rather than total angular momentum!



- **Degenerate orbitals:** $\phi_a(\mathbf{r}) \equiv \phi(\mathbf{r})$ and $\phi_b(\mathbf{r}) \equiv \phi(\mathbf{r} - \mathbf{a})$. Hence

$$\begin{aligned}
 \mathcal{J} &= (E_S - E_T)/2 \\
 &\approx \iint \frac{e^2}{4\pi\epsilon_0|\mathbf{r}_1 - \mathbf{r}_2|} \phi(\mathbf{r}_1)\phi(\mathbf{r}_1 - \mathbf{a})\phi(\mathbf{r}_2)\phi(\mathbf{r}_2 - \mathbf{a}) d^3\mathbf{r}_1 d^3\mathbf{r}_2 \\
 &\sim \pm \frac{e^2}{4\pi\epsilon_0 a_B} \left[\int \phi(\mathbf{r})\phi(\mathbf{r} - \mathbf{a}) d^3\mathbf{r} \right]^2 e^{-a/a_B} \sim \pm 1-100 \text{ meV}
 \end{aligned}$$

and can have either sign.

- **Crucial point:** energy difference between singlet & triplet spin configurations is $2\mathcal{J}$.

Interatomic Exchange in Solids (II)

- **Total spin:** $\hat{\mathbf{S}} = \hat{\mathbf{s}}_1 + \hat{\mathbf{s}}_2$.
 - Hence $\hat{\mathbf{S}}^2 = \hat{\mathbf{s}}_1^2 + \hat{\mathbf{s}}_2^2 + 2\hat{\mathbf{s}}_1 \cdot \hat{\mathbf{s}}_2$.
 - Also, $[\hat{\mathbf{s}}_1, \hat{\mathbf{s}}_1] = [\hat{\mathbf{s}}_1, \hat{\mathbf{s}}_2] = \mathbf{0}$, so $\hat{\mathbf{s}}_1^2$, $\hat{\mathbf{s}}_2^2$ and $\hat{\mathbf{S}}^2$ have eigenstates in common.
- Since $\hat{\mathbf{s}}^2|\psi\rangle = s(s+1)\hbar^2|\psi\rangle$ for an eigenstate $|\psi\rangle$ of $\hat{\mathbf{s}}^2$ it follows that

$$\hat{\mathbf{s}}_1 \cdot \hat{\mathbf{s}}_2|\psi\rangle = \frac{S(S+1) - s_1(s_1+1) - s_2(s_2+1)}{2}\hbar^2|\psi\rangle.$$

- For two spin- $\frac{1}{2}$ particles $s_1 = s_2 = 1/2$.
 - **Singlet:** $S = 0$ $\hat{\mathbf{s}}_1 \cdot \hat{\mathbf{s}}_2|\psi\rangle = \frac{1}{2} \left(0 \cdot 1 - \frac{3}{4} - \frac{3}{4} \right) \hbar^2|\psi\rangle = -\frac{3}{4}\hbar^2|\psi\rangle$.
 - **Triplet:** $S = 1$ $\hat{\mathbf{s}}_1 \cdot \hat{\mathbf{s}}_2|\psi\rangle = \frac{1}{2} \left(1 \cdot 2 - \frac{3}{4} - \frac{3}{4} \right) \hbar^2|\psi\rangle = \frac{1}{4}\hbar^2|\psi\rangle$.
- For two localised spins, interatomic exchange effects are described by the Hamiltonian $\hat{H} = -2\mathcal{J}\hat{\mathbf{s}}_1 \cdot \hat{\mathbf{s}}_2/\hbar^2$.
 - This gives $E_S - E_T = 2\mathcal{J}$ as expected. (Constant energy offset doesn't matter.)

Interatomic Exchange in Solids (III): Heisenberg Model

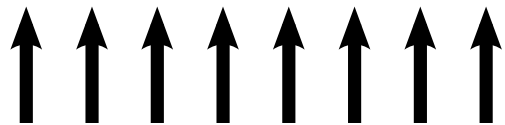
- Now consider the Hamiltonian for N spins on a lattice:

$$\hat{H} = -\frac{\mathcal{J}}{\hbar^2} \sum_{p=1}^N \sum_{\alpha} \hat{\mathbf{S}}_p \cdot \hat{\mathbf{S}}_{p+\alpha}$$

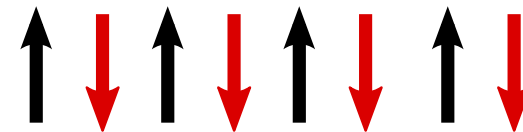
where $\sum_p$ runs over the N lattice sites and $\sum_{\alpha}$ runs over the z_c nearest neighbours of site p .

- N.b., we have included a factor of $1/2$ to eliminate double counting of pairs; this cancels the factor of 2 from the definition of $\mathcal{J}$.
- Beyond-nearest-neighbour interactions are neglected.

- At low T :



$\mathcal{J} > 0$: ferromagnetic order



$\mathcal{J} < 0$: antiferromagnetic order

Mean-Field Approximation for the FM Heisenberg Model (I)

- Hamiltonian for N spins with nearest-neighbour coupling and external field $\mathbf{B}$:

$$\hat{H} = -\frac{\mathcal{J}}{\hbar^2} \sum_{p=1}^N \sum_{\alpha} \hat{\mathbf{S}}_p \cdot \hat{\mathbf{S}}_{p+\alpha} + g_s \frac{\mu_B}{\hbar} \mathbf{B} \cdot \sum_{p=1}^N \hat{\mathbf{S}}_p.$$

- $\langle \dots \rangle =$ QM expectation value + thermal average.
- Connected correlation of spins:

$$\underbrace{\left(\hat{\mathbf{S}}_p - \langle \hat{\mathbf{S}} \rangle \right) \cdot \left(\hat{\mathbf{S}}_r - \langle \hat{\mathbf{S}} \rangle \right)}_{\text{“correlation of fluctuations”}} = \hat{\mathbf{S}}_p \cdot \hat{\mathbf{S}}_r - \left(\hat{\mathbf{S}}_p + \hat{\mathbf{S}}_r \right) \cdot \langle \hat{\mathbf{S}} \rangle + \underbrace{\langle \hat{\mathbf{S}} \rangle^2}_{\text{const.}}$$

- **Mean-field approximation:** *neglect connected correlations for $p \neq r$.* Then

$$\hat{\mathbf{S}}_p \cdot \hat{\mathbf{S}}_r \approx \left(\hat{\mathbf{S}}_p + \hat{\mathbf{S}}_r \right) \cdot \langle \hat{\mathbf{S}} \rangle - \langle \hat{\mathbf{S}} \rangle^2.$$

Mean-Field Approximation for the FM Heisenberg Model (II)

- Mean-field approximation to Hamiltonian:

$$\begin{aligned}\hat{H}_{\text{MF}} &= -\frac{2\mathcal{J}z_c}{\hbar^2} \sum_{p=1}^N \langle \hat{\mathbf{S}} \rangle \cdot \hat{\mathbf{S}}_p + \frac{\mathcal{J}Nz_c}{\hbar^2} \langle \hat{\mathbf{S}} \rangle^2 + g_s \frac{\mu_B}{\hbar} \mathbf{B} \cdot \sum_{p=1}^N \hat{\mathbf{S}}_p \\ &= \left(-\frac{2\mathcal{J}z_c}{\hbar^2} \langle \hat{\mathbf{S}} \rangle + g_s \frac{\mu_B}{\hbar} \mathbf{B} \right) \cdot \sum_{p=1}^N \hat{\mathbf{S}}_p + \text{const.} \equiv g_s \frac{\mu_B}{\hbar} \mathbf{B}_{\text{eff}} \cdot \sum_{p=1}^N \hat{\mathbf{S}}_p + \text{const.},\end{aligned}$$

where each spin moves **independently** in an effective magnetic field $\mathbf{B}_{\text{eff}} = \mathbf{B} - \frac{2\mathcal{J}z_c}{g_s\mu_B\hbar} \langle \hat{\mathbf{S}} \rangle$.

- Can assume that $\langle \hat{\mathbf{S}} \rangle$ is parallel to $\mathbf{B}$. As usual, assume that $\mathbf{B}_{\text{eff}} = B_{\text{eff},z} \mathbf{e}_z$.
- *Self-consistency*: calculate $\langle \hat{\mathbf{S}} \rangle$ using $\hat{H}_{\text{MF}}$.
- Mean-field Hamiltonian for a single spin: $\hat{\mathcal{H}}_{\text{MF}} = g_s\mu_B B_{\text{eff},z} \hat{S}_z / \hbar$.
 - We solved this problem before, when we looked at *Quantum Paramagnetism: Zeeman Splitting* in Part 1.

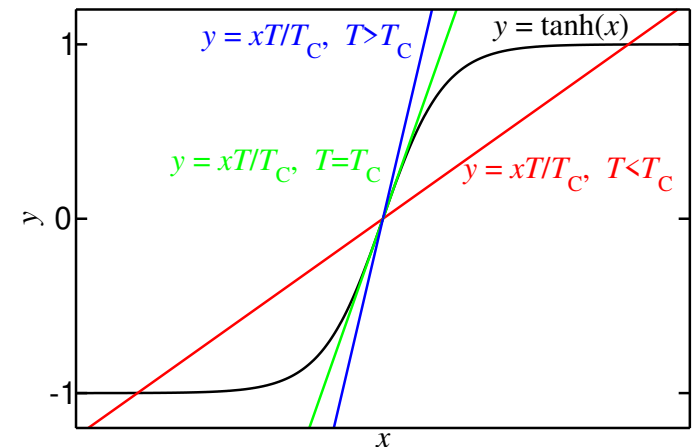
Mean-Field Approximation for the FM Heisenberg Model (III)

- Let's assume there is no external magnetic field, i.e., $\mathbf{B} = \mathbf{0}$. For simplicity also take $S = 1/2$.
 - For $S_z = \pm \hbar/2$, the single-particle energies are $\mp \mathcal{J} z_c \langle \hat{S}_z \rangle / \hbar$.
 - Using Boltzmann statistics [$\text{Pr}(U) \propto \exp(-\beta U)$, where $\beta = 1/(k_B T)$],

$$\langle \hat{S}_z \rangle = \frac{(\hbar/2)e^{\beta \mathcal{J} z_c \langle \hat{S}_z \rangle / \hbar} - (\hbar/2)e^{-\beta \mathcal{J} z_c \langle \hat{S}_z \rangle / \hbar}}{e^{\beta \mathcal{J} z_c \langle \hat{S}_z \rangle / \hbar} + e^{-\beta \mathcal{J} z_c \langle \hat{S}_z \rangle / \hbar}} = \frac{\hbar}{2} \tanh \left(\frac{z_c \mathcal{J}}{k_B T \hbar} \langle \hat{S}_z \rangle \right).$$

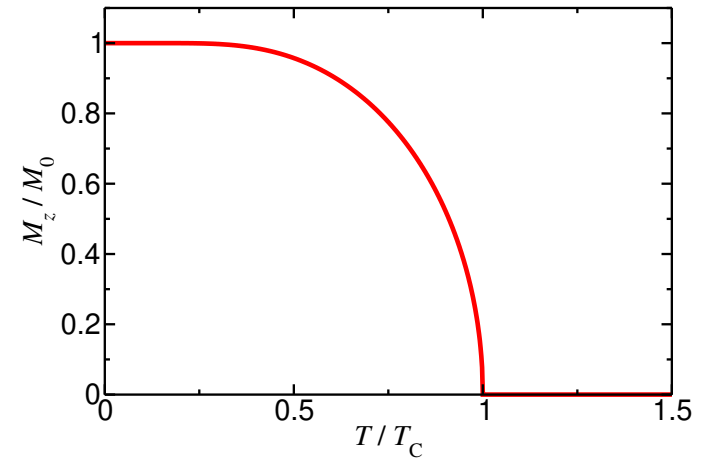
- Seek a **self-consistent** solution for $\langle \hat{S}_z \rangle$.
 - One solution is always $\langle \hat{S}_z \rangle = 0$.
 - May have nonzero solution, with lower energy.

- Let $x = \frac{z_c \mathcal{J}}{k_B T \hbar} \langle \hat{S}_z \rangle$. Then $\frac{2k_B T}{z_c \mathcal{J}} x = \tanh(x)$.
- Critical case [*Curie temperature* $T_C = z_c \mathcal{J} / (2k_B)$] is when gradient of left-hand-side is equal the gradient of the right-hand-side at $x = 0$.



Mean-Field Approximation for the FM Heisenberg Model (IV)

- At low temperature, $\langle \hat{S}_z \rangle$ saturates at $\pm \hbar/2$. At higher temperatures, $\langle \hat{S}_z \rangle$ falls to zero at temperature $T = T_C$.
- Magnetisation $M_z = -g_s \mu_B \langle \hat{S}_z \rangle / (\hbar V)$ behaves similarly; saturates at $M_0 = \pm g_s \mu_B / (2V)$.
- Limiting cases:
 - $T \rightarrow T_C^-$: $M_z(T) \propto (1 - T/T_C)^{1/2}$ (second-order phase transition).
 - $T \rightarrow 0$: $(M_0 - M_z)/M_0 \propto \exp(-2T_C/T)$.
- Experimentally: $\Delta M_z/M_0 \approx AT^{3/2}$ as $T \rightarrow 0$:
 - Ni: $A = (7.5 \pm 0.2) \times 10^{-6} \text{ K}^{-3/2}$,
 - Fe: $A = (3.4 \pm 0.2) \times 10^{-6} \text{ K}^{-3/2}$.



The reason for this: *spin waves*, a.k.a. *magnons* (next step).

Spin-Wave Excitations of a Ferromagnet (I)

“In the time of the first edition. . . only a few physicists then believed in the reality of spin waves” (C. Kittel, 1953)

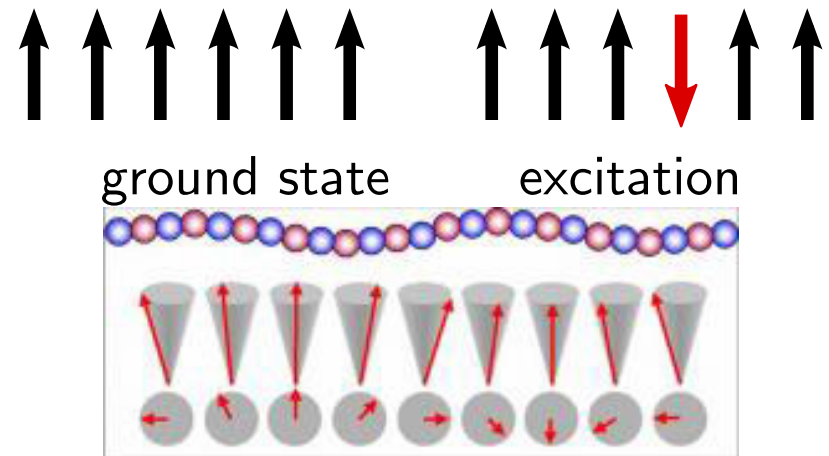
- One-dimensional chain of atoms subject to periodic boundary conditions:
 - **Hamiltonian**: $\hat{H} = -\frac{2\mathcal{J}}{\hbar^2} \sum_p \hat{\mathbf{S}}_p \cdot \hat{\mathbf{S}}_{p+1}$. (N.b., no double counting.)

- **Excitations**: changes in magnetic spin quantum number (spin flips for $S = 1/2$).

- Imagine slowly increasing coupling const. $\mathcal{J}$ from zero: excitation energy changes, but excitation still has one spin flip.

- **In ground state**: $\langle \mathbf{S}_p \cdot \mathbf{S}_{p+1} \rangle = S^2 \hbar^2$. So $E_0 = -2N\mathcal{J}S^2$.

- **Flip a single spin**: naïvely, energy would be $E_1 \stackrel{?}{=} E_0 + 8\mathcal{J}S^2$.
 - Can actually offset most of this energy loss by **delocalising** the excitation.
 - **Semiclassical picture**: treat atomic spins as classical angular-momentum vectors of length $\hbar S$.



Spin-Wave Excitations of a Ferromagnet (II)

- By Heisenberg equation of motion:

$$\begin{aligned} i\hbar \frac{d\langle \hat{S}_p^x \rangle}{dt} &= \langle [\hat{S}_p^x, \hat{H}] \rangle \\ &= -\frac{2\mathcal{J}}{\hbar^2} \langle [\hat{S}_p^x, \hat{\mathbf{S}}_p] \cdot (\hat{\mathbf{S}}_{p-1} + \hat{\mathbf{S}}_{p+1}) \rangle \\ &= -\frac{2\mathcal{J}}{\hbar^2} \langle (i\hbar \mathbf{e}_y \hat{S}_p^z - i\hbar \mathbf{e}_z \hat{S}_p^y) \cdot (\hat{\mathbf{S}}_{p-1} + \hat{\mathbf{S}}_{p+1}) \rangle \\ &= -\frac{2\mathcal{J}}{\hbar^2} \langle -i\hbar [\hat{\mathbf{S}}_p \times (\hat{\mathbf{S}}_{p-1} + \hat{\mathbf{S}}_{p+1})]_x \rangle. \end{aligned}$$

- Similarly for the y and z components (exercise). Hence

$$\frac{d\langle \hat{\mathbf{S}}_p \rangle}{dt} = \frac{2\mathcal{J}}{\hbar^2} \langle \hat{\mathbf{S}}_p \times (\hat{\mathbf{S}}_{p-1} + \hat{\mathbf{S}}_{p+1}) \rangle \approx \frac{2\mathcal{J}}{\hbar^2} \langle \hat{\mathbf{S}}_p \rangle \times \langle \hat{\mathbf{S}}_{p-1} + \hat{\mathbf{S}}_{p+1} \rangle,$$

where we have neglected correlation.

Spin-Wave Excitations of a Ferromagnet (III)

- Write $\mathbf{S}_p \equiv \langle \hat{\mathbf{S}}_p \rangle / \hbar$.
- Semiclassical equations of motion: $\hbar \frac{d}{dt} \mathbf{S}_p = 2\mathcal{J} \mathbf{S}_p \times (\mathbf{S}_{p+1} + \mathbf{S}_{p-1})$.
- These equations are nonlinear.
- Assuming small deviations from the ground state ($S_p^{x,y} \ll S$), *linearise* them: set $S_p^z = S$ and neglect terms containing $S^x S^y$ etc.
- Linearised equations of motion:

$$\begin{aligned}\frac{d}{dt} S_p^x &= \frac{2\mathcal{J}S}{\hbar} [2S_p^y - S_{p+1}^y - S_{p-1}^y] \\ \frac{d}{dt} S_p^y &= -\frac{2\mathcal{J}S}{\hbar} [2S_p^x - S_{p+1}^x - S_{p-1}^x] \\ \frac{d}{dt} S_p^z &= 0.\end{aligned}$$

Spin-Wave Excitations of a Ferromagnet (IV)

- Look for travelling wave solutions (can choose $\omega \geq 0$ without loss of generality):

$$S_p^x = C \exp [i(pka - \omega t)] \quad \text{and} \quad S_p^y = D \exp [i(pka - \omega t)] .$$

- This gives

$$-i\omega C = \frac{4\mathcal{J}S}{\hbar} [1 - \cos(ka)] D \quad \text{and} \quad -i\omega D = -\frac{4\mathcal{J}S}{\hbar} [1 - \cos(ka)] C,$$

i.e.,

$$\begin{pmatrix} i\omega & 4\mathcal{J}S[1 - \cos(ka)]/\hbar \\ -4\mathcal{J}S[1 - \cos(ka)]/\hbar & i\omega \end{pmatrix} \begin{pmatrix} C \\ D \end{pmatrix} = \mathbf{0}.$$

- A nonzero solution for C and D exists if and only if the determinant of the coefficient matrix is zero, which yields

$$\omega = 4\mathcal{J}S [1 - \cos(ka)] / \hbar.$$

- N.b., in the long-wavelength limit $ka \ll 1$, $\cos(ka) \approx 1 - (ka)^2/2$, so that $\omega \propto k^2$.

Spin-Wave Excitations of a Ferromagnet (V)

- In three dimensions, it is still the case that $\omega \propto k^2$ at small k .
- **Quantised spin waves:** *magnons*, elementary excitations on top of the ferromagnetic background.
 - For a mode $\mathbf{k}$ with $n_{\mathbf{k}}$ magnons, $\mathcal{E}_{\mathbf{k}} = (n_{\mathbf{k}} + 1/2)\hbar\omega_{\mathbf{k}}$.
 - Excitation of a magnon $\iff$ flipping of one spin.
 - Magnons are bosonic **quasiparticles** (spin-1).
 - In thermal equilibrium they have the Planck distribution (like photons or phonons):

$$n_{\mathbf{k}} = \frac{1}{e^{\hbar\omega_{\mathbf{k}}/(k_{\text{B}}T)} - 1} \equiv n_{\omega_{\mathbf{k}}}.$$

- Frequency density of states in 3D: $\mathcal{D}(\omega) \propto \sqrt{\omega}$.

Spin-Wave Excitations of a Ferromagnet (VI)

- Each magnon corresponds to flipping one spin:

$$\Delta M_z \propto \frac{N_{\text{magnons}}}{V} = \frac{1}{V} \sum_{\mathbf{k}} n_{\mathbf{k}} = \frac{1}{V} \int_0^\infty \mathcal{D}(\omega) n_\omega d\omega.$$

- Recall $\mathcal{D}(\omega) \propto \omega^{1/2}$ in $d = 3$ and $\omega_{\mathbf{k}} \propto k^2$. Hence

$$\begin{aligned} \Delta M_z &\propto \int_0^\infty \frac{\omega^{1/2} d\omega}{\exp(\hbar\omega/k_B T) - 1} \\ &\propto T^{3/2} \int_0^\infty \frac{x^{1/2} dx}{e^x - 1}, \end{aligned}$$

where $x = \hbar\omega/(k_B T)$.

- Bloch $T^{3/2}$ law:** $\Delta M_z \propto T^{3/2}$ in a ferromagnet at low temperature.
 - Can calculate prefactor: see C. Kittel, *Introduction to Solid State Physics*, 8th ed., chap. 12, section on magnons.
 - Experimental results in excellent agreement.

Neutron-Magnon Scattering

- Neutron scattering pattern for iron (taken from C. Kittel, *ISSP*):

$$\mathbf{q}_{\text{magnon}} = \mathbf{p} - \mathbf{p}' - \mathbf{G}$$
$$\hbar\omega_{\mathbf{q}} = \frac{\mathbf{p}^2}{2m} - \frac{\mathbf{p}'^2}{2m}.$$

- Magnetic moment of a neutron interacts with magnetic moments in a crystal.
- Hence scattering of neutrons off a magnetic crystal tells us about distribution, magnitude and direction of magnetic moments.

